University of Cape Town
Department of Computer Science

Computer Science CSC3003S
ASD Example Exam 2025

Approximate marks per question are shown in brackets.

READ ME FIRST:
This is a sample exam, not an actual exam.
If you are asked for n things and list more than n things, some of which are incorrect, you will receive minus points of any incorrect items. If you list n or fewer things you will not receive minus points for erroneous items.

This sample test has 5 pages.

Time: 2 hours + 10 mins reading time **Marks: 50**

Question 1 [10]

Consider the following pseudocode:

```
void CopyBuffer(int[] dest, int[] source, uint64 length) {
  for (uint64 i = 0; i < length; i++) {
    dest[i] = source[i];
  }
}
```

Assume that your programming language does not protect you; i.e. you are not using a 'safe' language.

a. Name and explain the security problem that lurks in this code. [2]

b. What problem can this security lapse cause? [2]

c. Either: rewrite the above code, or describe in prose what changes to make so that the above code is no longer unsecure. *Note that you may need to change the function's signature.* [3]

d. Why is using a 'secure' language like Rust (or Java) not enough to fix all security problems? [1]

e. Rust fixes the security problem shown in points a, b, c.
Which additional security problems does Rust fix?
either describe one (1) problem class, or name two (2) examples. [2]

Question 2 [6]

CPUs speed up execution using a number of mutually reinforcing mechanisms. One of these is pipelining.

a. Draw a diagram of a –simplified– CPU instruction pipeline, consisting of a handful of stages, showing multiple instruction cycles. Label the stages of the pipeline and explain briefly what each stage does. [3]

b. How does this pipelining speed up execution? [1]

c. What is out-of-order (OoO) execution and how does it help the pipeline? [2]

Question 3 [8]

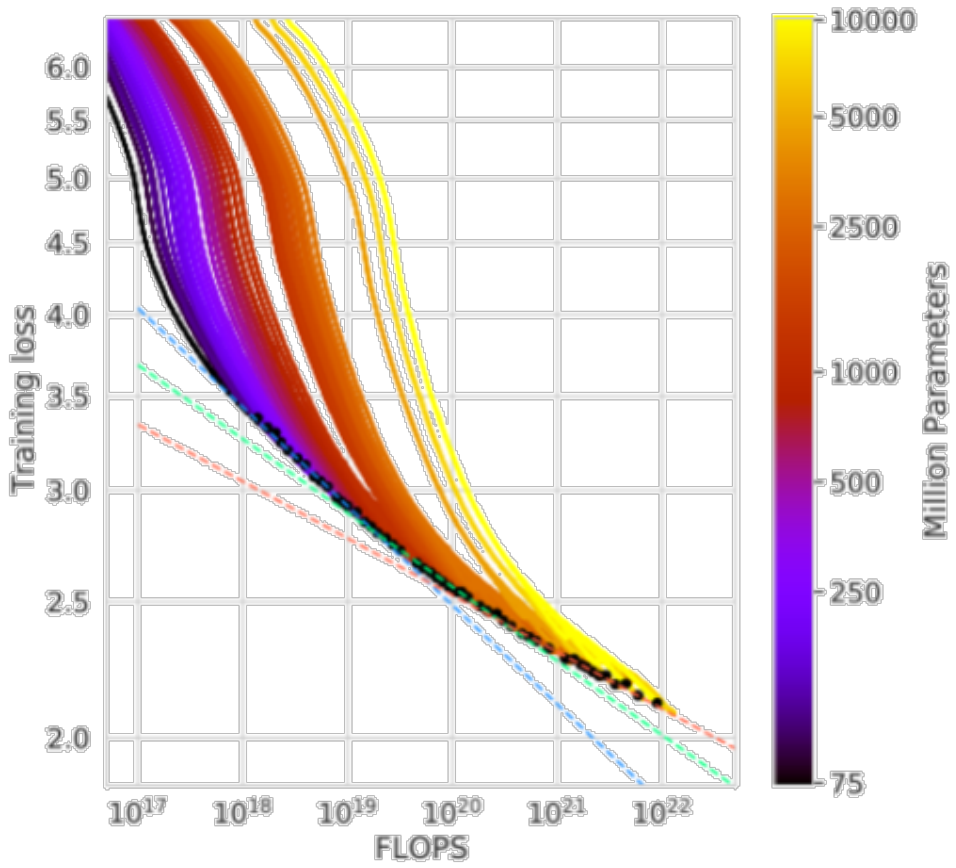
a. Xoliswa is a programmer that adheres to the clean coding principles. Name 3 things that she does in her code [3]

b. Describe the programming workflow used in test driven development [3]

c. *Xoliswa notices that her non-clean coding colleague also does not use unit tests and argues that such sloppy coding is a violation of professional ethics.* Put yourself in Xoliswa's shoes and using the ACM/IEEE code of ethics argue her point. [2]

Question 4 [4]

Consider the following graph detailing AI scaling.
Note that it is a log-log plot, i.e.: both axes scale logarithmically.
This graph shows that training loss (Y-axis) decreases as more FLoating point OPerationS (X-axis) are applied to the training of a generative AI (LLM or otherwise) model. It also shows that models with more parameters follow a different slope w.r.t. FLOPS vs loss. The dotted lines are fits over time (straight lines in this graph, curved lines on a graph with linear axes), blue being the oldest and red the newest fit. Assume that every year sees an increased number of parameters (Y1: black, Y2: purple, Y3: red, Y4: orange, Y5 yellow).



a. *Unsurprisingly, more compute (FLOPS) results in a lower training loss.*
Explain the **real** relationship this graph shows between applied compute power and training loss in mathematical terms. [2]

Considering that this graph lists actual achieved training losses using applied compute power, what does this graph say about:

b. AI scaling as a function of time? [1]

c. AI scaling as a function of applied compute power (FLOPS)? [1]

Question 5 [5]

For the "Pipe and filter architecture pattern" detail the following

a. Problem (what problem does this pattern solve) [1]

b. Solution (How does it solve it) [1]

c. Pros/Cons (Benefits + drawbacks using this pattern) [2]

d. Sketch the structure of the chosen pattern using a hard drawn diagram with label as needed. [1]

Question 6 [6]

a. The P in SOAP stands for 'protocol', what do the other letters stand for? [1]

b. SOAP is often used in combination with XML
What problem does this cause? [1]

c. Would you use SOAP in a micro-services architecture?
Why or why not? [2]

d. How does WSDL (web service description language) relate to SOAP? [2]

Question 7 [6]

Programmer Jane applies the "coding by exception antipattern.
Detail:

a. The problem (that this antipattern is erroneously applied to) [1]

b. The wrong solution (i.e.: the antipattern itself, the non-fix) [1]

c. Consequence (of implementing the antipattern) [1]

d. Fix (the correct solution to apply instead) [2]

Question 8 [5]

a. Explain the difference between unit testing and integration testing [2]

b. Complete the following table listing the pros and cons of aspect oriented programming (list 3 pros and 3 cons) [3]

Pros	Cons